

Review for Exam 1

1. Find the domain of the rational function $\frac{x-2}{x^2+3x+2}$

2. Completely simplify the following rational expressions.

(a) $\frac{x^3-9x}{x^2-3x}$

(b) $\frac{3x-12}{x^2-4} \cdot \frac{x^2+6x+8}{x-4}$

(c) $\frac{x^2+3x+2}{x+5} \div (x+1)$

(d) $\frac{11a^4b^3+22a^3b^4-55ab^5}{11ab^3}$

(Exam 1 review continued)

3. Add or subtract the rational expressions by first finding the LCD. Completely simplify your answers

(a) $\frac{1}{x+y} + \frac{3xy}{x^3+y^3}$

(b) $\frac{2}{x^2+5x+6} - \frac{4}{x^2+4x+3}$

(c) $\frac{4x}{x^2+6x+5} - \frac{3x}{x^2+5x+4} + \frac{1}{x+1}$

(Exam 1 review continued)

4. Solve the following rational equations. Don't forget to make note of any x values that cannot be solutions because they would cause you to divide by 0.

(a) $\frac{x+4}{x^2+5x} = \frac{-2}{x^2-25}$

(b) $1 - \frac{1}{x} = \frac{12}{x^2}$

5. Completely simplify the complex fractions by first finding the LCD of all “embedded” fractions.

(a) $\frac{\frac{3}{x^2} + \frac{1}{x}}{2 - \frac{4}{5x}}$

(b) $\frac{\frac{3}{x+2} - \frac{1}{x-2}}{\frac{4}{x^2-4}}$

(Exam 1 review continued)

6. Solve $\frac{1}{p} + \frac{1}{r} = \frac{2}{t}$ for p .

7. y varies directly with x . If $y = 4$ when $x = 2$, find y when $x = 10$.

8. y varies inversely with x . If $y = 10$ when $x = 6$, find x when $y = 12$.

9. Two runners, one averaging 8 MPH and the other averaging 10 MPH, run on a track for the same amount of time. If the slower runner runs 6 miles, how far does the faster runner run? (*Hint: Use $t = \frac{d}{r}$ and note that $t_1 = t_2$.*)

1.7(a) Scientific Notation

Write the following numbers in standard decimal notation.

1. 3.956×10^2

2. 2.39×10^{-4}

3. -6.052×10^{-6}

4. 1.001×10^5

Write the following numbers in scientific notation.

5. 84,060,000

6. 0.00012

7. -42.215

8. -0.00201

(1.7(a) continued)

9. The moon has an average distance from the Earth of approximately 2.389×10^5 miles. What is this distance in standard decimal notation?

10. One second is approximately 0.0000000317 years. Convert this number to scientific notation.

Review of exponents: Completely simplify the following expressions.

11. $(-6)^{-2}$

12. $\frac{1}{2^{-1}} + \frac{1}{3^{-1}}$

13. $(4b^{-9})(5b^4)$

14. $\frac{-28b^{-2}c^{-3}}{4b^{-3}c^{-1}}$

1.7(b) Operations with Scientific Notation

Perform the indicated operation. Give answers in scientific notation.

1. $(3.6 \times 10^8)(3 \times 10^2)$

2. $\frac{5.4 \times 10^3}{3 \times 10^6}$

3. $\frac{4.2 \times 10^5}{2.1 \times 10^{-2}}$

4. $(6 \times 10^{10})(2 \times 10^{-8})$

5. $(1.7 \times 10^{-6})(5.2 \times 10^{-4})$

6. $\frac{7.2 \times 10^{12}}{4 \times 10^3}$

5 Radicals and Fractional Exponents

- Monday:

- Topic: Radicals (7.1)
- Study skills: Exam reflection
- Worksheets p. 36-37
- Homework: 7.1

- Tuesday:

- Topic: Fractional Exponents (7.2)
- **Quiz:** 1.7(b), 7.1
- Worksheets p. 38
- Homework: 7.2

- Wednesday:

- Topic: Simplifying Radical Expressions (7.3)
- Group writing project (Ch. 7)
- Worksheets p. 39-40
- Homework: 7.3

- Thursday:

- Topic: Operations on Radical Expressions (7.4)
- **Quiz:** 7.2, 7.3
- Worksheets p. 41-42
- **Worksheets p. 35-42 due at end of class**
- Homework: 7.4

7.1 Radicals

Evaluate the following radicals or state that they do not exist.

1. $\sqrt{-16}$

2. $-\sqrt{16}$

3. $\sqrt[3]{-27}$

4. $-\sqrt[3]{27}$

5. $\sqrt[4]{81}$

6. $\sqrt[4]{-81}$

7. $\sqrt{-1}$

8. $\sqrt[3]{-1}$

(7.1 continued)

Answer the questions about the following radical functions.

9. $f(x) = \sqrt{15 - x}$

(a) Find $f(6)$

(b) Find $f(0)$

(c) Find $f(19)$

(d) Find the domain of $f(x)$

10. $g(x) = \sqrt{2x + 5}$

(a) Find $g(4)$

(b) Find $g(-3)$

(c) Find $g(-2)$

(d) Find the domain of $g(x)$